Supplementary material

QLoRA Fine-Tuning for Consumer Complaint Classification

Implementation, full results and reproducibility information

This supplement describes data allocation, model implementation, validation searches, complete results, input-length and latency conditions, and reference-parameter sensitivity. Auxiliary analyses specify their fitting populations and dependence limitations. The reproducibility package supplies the corresponding machine-readable data, predictions and verification code.

Contents

S1 Data provenance, fixed selections and visible-input checks **2**

S2 Implementation, immutable artifacts and validation searches **3**

S2.1 Pinned models and execution environment . . . 3

S2.2 QLoRA and inference . . . 4

S2.3 Encoder selection and fresh refits . . . 4

S2.4 Sparse baseline selection . . . 5

S3 Full model, seed, class and uncertainty results **6**

S4 Benchmark overlaps and per-view descriptions **11**

S5 Input-length conditions, cohorts and sensitivity analyses **12**

S5.1 Auxiliary length-stratum and matched-view analyses . . . 13

S6 Controlled latency and measurement limitations **14**

S7 Reference [14] encoder-parameter sensitivity **16**

S8 Reproducibility and auxiliary protocol analyses **17**

S8.1 Artifact map and verification . . . 17

S8.2 Auxiliary data-overlap analysis . . . 18

S8.3 Configuration screening for the shared QLoRA settings . . . 19

S8.4 Validation scope . . . 19

S8.5 Computational cost and CPU baseline timings . . . 19

S1 Data provenance, fixed selections and visible-input checks

The local source ZIP has 162,421 rows; the working file retains its first 162,415 rows identically. The six absent trailing records have the placeholder narrative `name`. Ten records with missing required fields are removed, leaving 162,405 usable rows. Excluding 8,123 benchmark-related source rows produces a 154,282-row partitioning pool. Model-visible overlap screening excludes 2,925 candidate training and 307 candidate validation rows from fitting while retaining the evaluation records. These operations document the local archive, not the unavailable original CFPB-to-Kaggle preprocessing or product aggregation.

The benchmark-related exclusion records comprise 4,990 exact normalised-text matches, 3,096 250-character prefix matches, 23 additional narrative-group members and 14 cross-label prefix matches. These counts describe source-pool filtering; model-visible screening additionally excludes fitting conflicts under the actual input representations.

Labels are inherited by identity. Their code order is retail_banking, credit_reporting, credit_card, mortgages_and_loans, debt_collection. Connected components use full normalisation, 500-character and reconstructed tokenizer representations, with a conservative 250-character benchmark prefix guard. They represent text dependencies, not verified complaint IDs. No semantic/paraphrase independence guarantee follows.

Table S1: Retained and actually selected class counts. The full training pool is used only by the full-data sparse baseline.

Class	Train pool	Val. pool	Internal	Train 10k	Val. 2k
Retail banking	10,342	1,293	1,293	858	171
Credit reporting	65,750	8,267	8,571	5,456	1,093
Credit card	11,891	1,524	1,494	987	202
Mortgages/loans	14,696	1,837	1,826	1,220	243
Debt collection	17,823	2,199	2,244	1,479	291

Table S2: Candidate allocation, excluded fitting conflicts, retained rows and conservative component counts.

Role	Candidate rows	Excluded	Retained	Groups
Train	123,427	2,925	120,502	93,609
Validation	15,427	307	15,120	11,925
Internal	15,428	0	15,428	12,099

Table S3: Actual pinned-tokenizer audits. Every model has 10k train + 2k validation + 15,428 internal + 12k benchmark appearances. Zero refers only to the tested train/validation split intersections.

Model	Inputs	Separation failures	Passed
qwen	39428	0	True
llama	39428	0	True
bert	39428	0	True
distilbert	39428	0	True
roberta	39428	0	True

The 25 retained-pool representation checks validate the allocation, while the five pinned-tokenizer audits verify the exact selected inputs supplied to the models. Longer-input keys and both reference-parameter encoder tokenizers are checked separately. Equal token inputs remain inside the frozen bootstrap groups. All data hashes and row selections are in `analysis/revision_v1/protocol.json` and per-run manifests.

The internal cohort is fixed and inspected, so evaluation is retrospective. Benchmark files are byte-identical to source-ZIP members. Their sampling implementation is unavailable; seed identifiers and balanced/imbalanced/length conditions are documented from the frozen files. The local source derivation does not recover unknown upstream transformations.

S2 Implementation, immutable artifacts and validation searches

S2.1 Pinned models and execution environment

All neural runs use locally cached, immutable Hugging Face snapshots. Weight files were checked against the pinned Hub metadata by full hashes, sizes and safetensors headers/offsets. The identities in Table S4 define the primary-comparison models.

Table S4: Immutable model identities used for the primary comparison.

Model	Repository	Commit
qwen	`Qwen/Qwen3-8B`	`b968826d9c46dd6066d109eabc6255188de91218`
llama	`meta-llama/Llama-3.1-8B-Instruct`	`0e9e39f249a16976918f6564b8830bc894c89659`
bert	`google-bert/bert-base-uncased`	`86b5e0934494bd15c9632b12f734a8a67f723594`
distilbert	`distilbert/distilbert-base-uncased`	`12040accade4e8a0f71eabdb258fecc2e7e948be`
roberta	`FacebookAI/roberta-base`	`e2da8e2f811d1448a5b465c236feacd80ffbac7b`

Table S5: Actual GPU environment. Complete pip freeze and code snapshots are retained per job. CPU statistics use the separately recorded analysis environment (pandas 3.0.3; scikit-learn 1.9.0).

Component	Value
GPU	NVIDIA RTX 5060 Ti 16 GB; driver 595.99.02
Python	3.12.14
PyTorch/CUDA	2.7.1+cu128 / 12.8
transformers	4.57.6
trl	0.23.1
peft	0.17.1
bitsandbytes	0.46.1
datasets	4.0.0
accelerate	1.10.1
safetensors	0.5.3
tokenizers	0.22.2
huggingface_hub	0.36.2
numpy	2.2.6
pandas	2.2.3
scikit-learn	1.6.1

S2.2 QLoRA and inference

Table S6: QLoRA and generation settings.

Setting	Value
Training / validation	10,000 / 2,000 fixed rows; selection seed 42
Optimisation seeds	42, 43, 44; fresh same-data refits
Quantisation	4-bit NF4, double quantisation, FP16 compute
Adapter	Rank 16; alpha 32; dropout 0.05; all-linear targets
Objective	Completion-only cross-entropy over target JSON and EOS
Epochs / LR	2 / 0.0002
Physical / effective batch	1 / 16 (accumulation 16)
Optimiser / scheduler	Paged AdamW 8-bit / cosine
Warmup	0.03
Gradient checkpointing	Non-reentrant
Model selection	Best saved validation loss, every 100 optimiser steps
Main visible narrative	First 500 characters; input cap 512 tokens
Generation	Greedy; one beam; maximum 32 new tokens; cache enabled
Thinking / serialization	`enable_thinking=False`; render once; no added duplicate BOS
Output handling	Final product-field regex; invalid or absent labels remain errors; no retries

The exact system prompt below is taken from the retained evaluation code; the user message is `Complaint:` followed by a newline and the truncated narrative. The target is `{"product": "label"}` followed by EOS. The template boundary and the full target fitting the 512-token SFT limit are verified for every selected training/validation record. JSON-format prompting is not token-level constrained decoding.

```
You are an AI assistant specializing in consumer complaint classification. Classify each complaint
↪ into exactly one of these categories: ['retail_banking', 'credit_reporting', 'credit_card',
↪ 'mortgages_and_loans', 'debt_collection']. Return only valid JSON in this format: {"product":
↪ "chosen_category_name"}. Do not provide explanations.
```

Displayed prompt line wraps are typographical; the source string and rendered templates in the retained code determine actual serialization. Exact-input memoisation is used only in deterministic LLM prediction generation. Identity includes token IDs/masks, immutable base and fitted-weight hashes, generation settings, runner/cache code hashes, library versions, hardware and driver. All appearances, labels and row metadata remain in prediction tables. Labels and row identities are not cache keys. Latency and warmups always make fresh GPU calls. The two 500-character token-cap control settings have identical encoded inputs and may reuse replies; they are not independent inference replicates.

S2.3 Encoder selection and fresh refits

Main BERT, DistilBERT and RoBERTa use 500 characters and a 512-token cap, physical batch 4/accumulation 4, evaluation batch 16, FP16, AdamW, weight decay 0.01, linear scheduling and warmup 0.06. Search seed is 42. Each candidate is evaluated at its final epoch; the globally highest validation macro-F1 wins, with the smallest candidate ID for ties within 10^{-12}. The selected DistilBERT and RoBERTa conditions are at the upper tested learning rate (5×10^{-5}) and epoch count (three). Scores outside that grid are unknown; the search is not a global-optimisation claim. Three fresh refits use the selected configuration. Test inference is blocked unless all 2,000 reloaded validation predictions exactly match their Trainer predictions.

Table S7: All 18 main encoder validation candidates. Only validation enters selection.

Model	ID	LR	Epochs	Val. macro-F1	Decision
bert	1	2e-05	2	0.7971	
bert	2	2e-05	3	0.8083	Selected
bert	3	3e-05	2	0.8019	
bert	4	3e-05	3	0.7996	
bert	5	5e-05	2	0.7937	
bert	6	5e-05	3	0.7988	

Table S7 continued

Model	ID	LR	Epochs	Val. macro-F1	Decision
distilbert	1	2e-05	2	0.7846	
distilbert	2	2e-05	3	0.7880	
distilbert	3	3e-05	2	0.7890	
distilbert	4	3e-05	3	0.7943	
distilbert	5	5e-05	2	0.7874	
distilbert	6	5e-05	3	0.8108	Selected
roberta	1	2e-05	2	0.7876	
roberta	2	2e-05	3	0.8020	
roberta	3	3e-05	2	0.7915	
roberta	4	3e-05	3	0.7997	
roberta	5	5e-05	2	0.7895	
roberta	6	5e-05	3	0.8108	Selected

S2.4 Sparse baseline selection

Each budget has 48 candidates. Fixed settings are lowercasing, Unicode accent stripping, sublinear term frequency, L2 normalisation, lbfgs, tolerance 10^{-5} and maximum 2,000 iterations. Vocabulary/IDF fit training only. Validation and test transformations never refit the feature map. The highest validation macro-F1 wins with the same predefined smallest-ID tie rule. The seed-labelled lbfgs fits have identical predictions at each budget; their SD zero is a determinism check, not stochastic seed evidence.

Table S8: All 96 sparse validation candidates (48 per budget). An asterisk marks the selected candidate; full means 120,502 training rows.

Budget	ID	N-grams	Max feat.	Min df	C	Weight	Val. mF1	
10000	0	(1,1)	50,000	2	0.25	none	0.7695	
10000	1	(1,1)	50,000	2	0.25	balanced	0.7560	
10000	2	(1,1)	50,000	2	1	none	0.7962	
10000	3	(1,1)	50,000	2	1	balanced	0.7685	
10000	4	(1,1)	50,000	2	4	none	0.7834	
10000	5	(1,1)	50,000	2	4	balanced	0.7723	
10000	6	(1,1)	50,000	5	0.25	none	0.7681	
10000	7	(1,1)	50,000	5	0.25	balanced	0.7599	
10000	8	(1,1)	50,000	5	1	none	0.7933	
10000	9	(1,1)	50,000	5	1	balanced	0.7671	
10000	10	(1,1)	50,000	5	4	none	0.7843	
10000	11	(1,1)	50,000	5	4	balanced	0.7688	
10000	12	(1,1)	200,000	2	0.25	none	0.7695	
10000	13	(1,1)	200,000	2	0.25	balanced	0.7560	
10000	14	(1,1)	200,000	2	1	none	0.7962	
10000	15	(1,1)	200,000	2	1	balanced	0.7685	
10000	16	(1,1)	200,000	2	4	none	0.7834	
10000	17	(1,1)	200,000	2	4	balanced	0.7723	
10000	18	(1,1)	200,000	5	0.25	none	0.7681	
10000	19	(1,1)	200,000	5	0.25	balanced	0.7599	
10000	20	(1,1)	200,000	5	1	none	0.7933	
10000	21	(1,1)	200,000	5	1	balanced	0.7671	
10000	22	(1,1)	200,000	5	4	none	0.7843	
10000	23	(1,1)	200,000	5	4	balanced	0.7688	
10000	24	(1,2)	50,000	2	0.25	none	0.7505	
10000	25	(1,2)	50,000	2	0.25	balanced	0.7582	
10000	26	(1,2)	50,000	2	1	none	0.7859	
10000	27	(1,2)	50,000	2	1	balanced	0.7735	
10000	28	(1,2)	50,000	2	4	none	0.7987	
10000	29	(1,2)	50,000	2	4	balanced	0.7833	
10000	30	(1,2)	50,000	5	0.25	none	0.7631	
10000	31	(1,2)	50,000	5	0.25	balanced	0.7643	
10000	32	(1,2)	50,000	5	1	none	0.7906	
10000	33	(1,2)	50,000	5	1	balanced	0.7733	
10000	34	(1,2)	50,000	5	4	none	0.7989	*
10000	35	(1,2)	50,000	5	4	balanced	0.7815	
10000	36	(1,2)	200,000	2	0.25	none	0.7494	
10000	37	(1,2)	200,000	2	0.25	balanced	0.7578	
10000	38	(1,2)	200,000	2	1	none	0.7869	
10000	39	(1,2)	200,000	2	1	balanced	0.7739	

Table S8 continued

Budget	ID	N-grams	Max feat.	Min df	C	Weight	Val. mF1	
10000	40	(1,2)	200,000	2	4	none	0.7985	
10000	41	(1,2)	200,000	2	4	balanced	0.7853	
10000	42	(1,2)	200,000	5	0.25	none	0.7631	
10000	43	(1,2)	200,000	5	0.25	balanced	0.7643	
10000	44	(1,2)	200,000	5	1	none	0.7906	
10000	45	(1,2)	200,000	5	1	balanced	0.7733	
10000	46	(1,2)	200,000	5	4	none	0.7989	
10000	47	(1,2)	200,000	5	4	balanced	0.7815	
full	0	(1,1)	50,000	2	0.25	none	0.8189	
full	1	(1,1)	50,000	2	0.25	balanced	0.7859	
full	2	(1,1)	50,000	2	1	none	0.8227	
full	3	(1,1)	50,000	2	1	balanced	0.7905	
full	4	(1,1)	50,000	2	4	none	0.8099	
full	5	(1,1)	50,000	2	4	balanced	0.7866	
full	6	(1,1)	50,000	5	0.25	none	0.8183	
full	7	(1,1)	50,000	5	0.25	balanced	0.7871	
full	8	(1,1)	50,000	5	1	none	0.8228	
full	9	(1,1)	50,000	5	1	balanced	0.7877	
full	10	(1,1)	50,000	5	4	none	0.8100	
full	11	(1,1)	50,000	5	4	balanced	0.7853	
full	12	(1,1)	200,000	2	0.25	none	0.8189	
full	13	(1,1)	200,000	2	0.25	balanced	0.7859	
full	14	(1,1)	200,000	2	1	none	0.8227	
full	15	(1,1)	200,000	2	1	balanced	0.7905	
full	16	(1,1)	200,000	2	4	none	0.8099	
full	17	(1,1)	200,000	2	4	balanced	0.7866	
full	18	(1,1)	200,000	5	0.25	none	0.8183	
full	19	(1,1)	200,000	5	0.25	balanced	0.7871	
full	20	(1,1)	200,000	5	1	none	0.8228	
full	21	(1,1)	200,000	5	1	balanced	0.7877	
full	22	(1,1)	200,000	5	4	none	0.8100	
full	23	(1,1)	200,000	5	4	balanced	0.7853	
full	24	(1,2)	50,000	2	0.25	none	0.8247	
full	25	(1,2)	50,000	2	0.25	balanced	0.7866	
full	26	(1,2)	50,000	2	1	none	0.8325	
full	27	(1,2)	50,000	2	1	balanced	0.8077	
full	28	(1,2)	50,000	2	4	none	0.8272	
full	29	(1,2)	50,000	2	4	balanced	0.8187	
full	30	(1,2)	50,000	5	0.25	none	0.8247	
full	31	(1,2)	50,000	5	0.25	balanced	0.7866	
full	32	(1,2)	50,000	5	1	none	0.8325	
full	33	(1,2)	50,000	5	1	balanced	0.8071	
full	34	(1,2)	50,000	5	4	none	0.8267	
full	35	(1,2)	50,000	5	4	balanced	0.8168	
full	36	(1,2)	200,000	2	0.25	none	0.8225	
full	37	(1,2)	200,000	2	0.25	balanced	0.7866	
full	38	(1,2)	200,000	2	1	none	0.8324	
full	39	(1,2)	200,000	2	1	balanced	0.8104	
full	40	(1,2)	200,000	2	4	none	0.8329	
full	41	(1,2)	200,000	2	4	balanced	0.8228	
full	42	(1,2)	200,000	5	0.25	none	0.8241	
full	43	(1,2)	200,000	5	0.25	balanced	0.7874	
full	44	(1,2)	200,000	5	1	none	0.8306	
full	45	(1,2)	200,000	5	1	balanced	0.8097	
full	46	(1,2)	200,000	5	4	none	0.8363	*
full	47	(1,2)	200,000	5	4	balanced	0.8258	

S3 Full model, seed, class and uncertainty results

All primary estimates use the same fixed training and validation selections. For adapted/encoder systems, points are means of three individually scored refits, not ensembles. SD is sample SD over those refits. Base systems have one checkpoint. Sparse repeated fits are deterministic. Intervals use 10,000 paired whole-group bootstrap draws (seed 20261020) and condition on fitted models/selection; they exclude training and hyperparameter-selection uncertainty. No family-wise adjustment is applied. Benchmark estimands equally weight unique visible label–narrative units rather than the 12,000 repeated appearances. All 19 main variants share each main-study draw.

For class k, F1 is $2TP_k/(2TP_k+FP_k+FN_k)$ with zero-denominator scores set to zero; macro-F1 averages the

five class scores and weighted F1 uses true-class support. The internal group-confusion tensor has five true classes and six prediction outcomes, including off-label. Off-label outputs therefore remain false negatives. Point estimates are computed from observed group counts before resampling. For computational efficiency, groups with identical joint confusion profiles across the analysed variants are collapsed into multinomial categories, with category probabilities equal to their frequencies divided by the group count. Sampling these categories preserves exactly the ordinary whole-group bootstrap distribution. The separate 12-variant reference analysis has a different joint-profile compression from the 19-variant main analysis; the same RNG seed does not imply identical Monte Carlo draws between those analyses. Within each analysis, every contrast still uses common paired draws.

More explicitly, let C_{gs} be the weighted confusion counts of group g for fitted variant s, and h the metric. A three-refit system has observed point $\widehat{\theta} = \frac{1}{3}\sum_{s=1}^{3} h(\sum_g C_{gs})$. In draw b, $(N_{1b}, \ldots, N_{Gb}) \sim \text{Multinomial}(G; 1/G, \ldots, 1/G)$ replaces each group count by $N_{gb}C_{gs}$. Paired differences use the same N_{gb} for both systems; the 2.5th and 97.5th percentiles define the interval. Base and deterministic sparse systems have one scored variant. Internal row weights are one. A benchmark appearance has weight $1/m_u$, where m_u is the number of appearances of its normalised label–narrative unit u; those weights remain fixed during resampling. Metrics are averaged after scoring each fitted variant, rather than scoring an averaged confusion matrix. The fitted checkpoints and selected configurations are held fixed throughout.

Table S9: Internal: 15,428 rows / 12,099 groups. All three metrics use observed points and common paired-group draws.

System	Metric	Point	95% CI	Seed SD
Qwen3 base	accuracy	0.7352	[0.7190, 0.7522]	—
Qwen3 base	macro_f1	0.6295	[0.6182, 0.6405]	—
Qwen3 base	weighted_f1	0.7206	[0.7040, 0.7380]	—
Qwen3 QLoRA	accuracy	0.8689	[0.8595, 0.8783]	0.0032
Qwen3 QLoRA	macro_f1	0.8331	[0.8245, 0.8411]	0.0048
Qwen3 QLoRA	weighted_f1	0.8678	[0.8583, 0.8772]	0.0034
Llama 3.1 base	accuracy	0.7295	[0.7130, 0.7467]	—
Llama 3.1 base	macro_f1	0.6247	[0.6138, 0.6349]	—
Llama 3.1 base	weighted_f1	0.7177	[0.7006, 0.7357]	—
Llama 3.1 QLoRA	accuracy	0.8736	[0.8645, 0.8828]	0.0007
Llama 3.1 QLoRA	macro_f1	0.8402	[0.8312, 0.8483]	0.0020
Llama 3.1 QLoRA	weighted_f1	0.8729	[0.8637, 0.8821]	0.0009
BERT	accuracy	0.8541	[0.8441, 0.8644]	0.0005
BERT	macro_f1	0.8119	[0.8028, 0.8203]	0.0010
BERT	weighted_f1	0.8528	[0.8427, 0.8632]	0.0004
DistilBERT	accuracy	0.8522	[0.8422, 0.8625]	0.0040
DistilBERT	macro_f1	0.8104	[0.8014, 0.8188]	0.0065
DistilBERT	weighted_f1	0.8511	[0.8411, 0.8614]	0.0043
RoBERTa	accuracy	0.8525	[0.8425, 0.8627]	0.0027
RoBERTa	macro_f1	0.8105	[0.8015, 0.8187]	0.0039
RoBERTa	weighted_f1	0.8520	[0.8419, 0.8622]	0.0030
TF-IDF/LR 10k	accuracy	0.8499	[0.8395, 0.8604]	0.0000
TF-IDF/LR 10k	macro_f1	0.8068	[0.7973, 0.8157]	0.0000
TF-IDF/LR 10k	weighted_f1	0.8474	[0.8370, 0.8581]	0.0000
TF-IDF/LR full	accuracy	0.8805	[0.8719, 0.8892]	0.0000
TF-IDF/LR full	macro_f1	0.8476	[0.8402, 0.8549]	0.0000
TF-IDF/LR full	weighted_f1	0.8798	[0.8712, 0.8885]	0.0000

Table S10: Benchmark: 5,909 weighted units / 3,919 groups. All three metrics use observed points and common paired-group draws.

System	Metric	Point	95% CI	Seed SD
Qwen3 base	accuracy	0.6785	[0.6633, 0.6936]	—
Qwen3 base	macro_f1	0.6324	[0.6151, 0.6491]	—
Qwen3 base	weighted_f1	0.6637	[0.6473, 0.6799]	—
Qwen3 QLoRA	accuracy	0.8304	[0.8191, 0.8415]	0.0025
Qwen3 QLoRA	macro_f1	0.8210	[0.8088, 0.8329]	0.0032
Qwen3 QLoRA	weighted_f1	0.8289	[0.8174, 0.8403]	0.0027

Table S10 continued

System	Metric	Point	95% CI	Seed SD
Llama 3.1 base	accuracy	0.6803	[0.6650, 0.6953]	—
Llama 3.1 base	macro_f1	0.6271	[0.6097, 0.6439]	—
Llama 3.1 base	weighted_f1	0.6623	[0.6456, 0.6787]	—
Llama 3.1 QLoRA	accuracy	0.8348	[0.8236, 0.8458]	0.0010
Llama 3.1 QLoRA	macro_f1	0.8272	[0.8152, 0.8390]	0.0009
Llama 3.1 QLoRA	weighted_f1	0.8336	[0.8223, 0.8448]	0.0009
BERT	accuracy	0.8076	[0.7959, 0.8194]	0.0036
BERT	macro_f1	0.7936	[0.7808, 0.8064]	0.0042
BERT	weighted_f1	0.8057	[0.7938, 0.8178]	0.0041
DistilBERT	accuracy	0.8038	[0.7921, 0.8158]	0.0061
DistilBERT	macro_f1	0.7894	[0.7767, 0.8021]	0.0069
DistilBERT	weighted_f1	0.8018	[0.7898, 0.8139]	0.0063
RoBERTa	accuracy	0.8063	[0.7946, 0.8180]	0.0037
RoBERTa	macro_f1	0.7926	[0.7798, 0.8051]	0.0042
RoBERTa	weighted_f1	0.8050	[0.7932, 0.8169]	0.0036
TF-IDF/LR 10k	accuracy	0.8001	[0.7871, 0.8128]	0.0000
TF-IDF/LR 10k	macro_f1	0.7871	[0.7728, 0.8006]	0.0000
TF-IDF/LR 10k	weighted_f1	0.7971	[0.7838, 0.8101]	0.0000
TF-IDF/LR full	accuracy	0.8450	[0.8335, 0.8562]	0.0000
TF-IDF/LR full	macro_f1	0.8347	[0.8220, 0.8467]	0.0000
TF-IDF/LR full	weighted_f1	0.8436	[0.8319, 0.8550]	0.0000

Table S11: Individual internal checkpoint scores. Each TF-IDF row represents three identical deterministic fits.

Variant	Accuracy	Macro-F1	Weighted F1
bert_42	0.8546	0.8130	0.8532
bert_43	0.8538	0.8110	0.8524
bert_44	0.8538	0.8117	0.8527
distilbert_42	0.8561	0.8168	0.8551
distilbert_43	0.8481	0.8037	0.8466
distilbert_44	0.8524	0.8107	0.8516
llama_42	0.8743	0.8424	0.8738
llama_43	0.8729	0.8384	0.8721
llama_44	0.8736	0.8397	0.8728
llama_base	0.7295	0.6247	0.7177
qwen_42	0.8717	0.8378	0.8709
qwen_43	0.8696	0.8332	0.8682
qwen_44	0.8654	0.8282	0.8642
qwen_base	0.7352	0.6295	0.7206
roberta_42	0.8545	0.8127	0.8538
roberta_43	0.8494	0.8060	0.8484
roberta_44	0.8537	0.8127	0.8536
tfidf_10000	0.8499	0.8068	0.8474
tfidf_full	0.8805	0.8476	0.8798

Table S12: Individual benchmark checkpoint scores. Each TF-IDF row represents three identical deterministic fits.

Variant	Accuracy	Macro-F1	Weighted F1
bert_42	0.8076	0.7935	0.8055
bert_43	0.8040	0.7894	0.8018
bert_44	0.8113	0.7979	0.8100
distilbert_42	0.8069	0.7926	0.8050
distilbert_43	0.7968	0.7814	0.7945
distilbert_44	0.8078	0.7941	0.8059
llama_42	0.8355	0.8282	0.8344

Table S12 continued

Variant	Accuracy	Macro-F1	Weighted F1
llama_43	0.8353	0.8269	0.8339
llama_44	0.8336	0.8265	0.8326
llama_base	0.6803	0.6271	0.6623
qwen_42	0.8289	0.8197	0.8274
qwen_43	0.8289	0.8187	0.8273
qwen_44	0.8333	0.8246	0.8321
qwen_base	0.6785	0.6324	0.6637
roberta_42	0.8096	0.7967	0.8081
roberta_43	0.8023	0.7883	0.8010
roberta_44	0.8069	0.7928	0.8059
tfidf_10000	0.8001	0.7871	0.7971
tfidf_full	0.8450	0.8347	0.8436

Table S13: All main macro-F1 paired contrasts, comparison minus reference. Full-data TF-IDF comparisons have unequal training budgets. The internal Llama-versus-full-data upper bound is $+6.8 \times 10^{-6}$ before rounding, so that interval includes zero.

View	Reference	Comparison	Difference	95% CI
internal	Qwen3 base	Qwen3 QLoRA	0.2035	[0.1934, 0.2141]
internal	Llama 3.1 base	Llama 3.1 QLoRA	0.2155	[0.2054, 0.2257]
internal	TF-IDF/LR 10k	Qwen3 QLoRA	0.0263	[0.0199, 0.0327]
internal	TF-IDF/LR 10k	Llama 3.1 QLoRA	0.0334	[0.0270, 0.0397]
internal	TF-IDF/LR 10k	BERT	0.0051	[-0.0013, 0.0117]
internal	TF-IDF/LR 10k	DistilBERT	0.0036	[-0.0025, 0.0098]
internal	TF-IDF/LR 10k	RoBERTa	0.0037	[-0.0025, 0.0098]
internal	TF-IDF/LR full	Qwen3 QLoRA	-0.0145	[-0.0221, -0.0071]
internal	TF-IDF/LR full	Llama 3.1 QLoRA	-0.0074	[-0.0153, 0.0000]
benchmark	Qwen3 base	Qwen3 QLoRA	0.1886	[0.1723, 0.2051]
benchmark	Llama 3.1 base	Llama 3.1 QLoRA	0.2000	[0.1838, 0.2166]
benchmark	TF-IDF/LR 10k	Qwen3 QLoRA	0.0340	[0.0230, 0.0452]
benchmark	TF-IDF/LR 10k	Llama 3.1 QLoRA	0.0401	[0.0296, 0.0511]
benchmark	TF-IDF/LR 10k	BERT	0.0065	[-0.0040, 0.0175]
benchmark	TF-IDF/LR 10k	DistilBERT	0.0023	[-0.0080, 0.0128]
benchmark	TF-IDF/LR 10k	RoBERTa	0.0056	[-0.0046, 0.0161]
benchmark	TF-IDF/LR full	Qwen3 QLoRA	-0.0136	[-0.0239, -0.0032]
benchmark	TF-IDF/LR full	Llama 3.1 QLoRA	-0.0075	[-0.0176, 0.0030]
internal	BERT	Qwen3 QLoRA	0.0212	[0.0160, 0.0264]
internal	BERT	Llama 3.1 QLoRA	0.0283	[0.0230, 0.0335]
internal	DistilBERT	Qwen3 QLoRA	0.0227	[0.0175, 0.0278]
internal	DistilBERT	Llama 3.1 QLoRA	0.0297	[0.0246, 0.0350]
internal	RoBERTa	Qwen3 QLoRA	0.0226	[0.0175, 0.0276]
internal	RoBERTa	Llama 3.1 QLoRA	0.0297	[0.0246, 0.0347]
benchmark	BERT	Qwen3 QLoRA	0.0274	[0.0181, 0.0364]
benchmark	BERT	Llama 3.1 QLoRA	0.0336	[0.0247, 0.0425]
benchmark	DistilBERT	Qwen3 QLoRA	0.0317	[0.0227, 0.0408]
benchmark	DistilBERT	Llama 3.1 QLoRA	0.0378	[0.0293, 0.0466]
benchmark	RoBERTa	Qwen3 QLoRA	0.0284	[0.0197, 0.0374]
benchmark	RoBERTa	Llama 3.1 QLoRA	0.0346	[0.0261, 0.0432]

Table S14: Internal class metrics. Multi-seed rows average the individually computed class statistics; supports are fixed.

System	Class	Precision	Recall	F1	Support
Qwen3 base	Retail banking	0.7762	0.4238	0.5483	1,293
Qwen3 base	Credit reporting	0.7924	0.8922	0.8394	8,571
Qwen3 base	Credit card	0.4760	0.7637	0.5865	1,494
Qwen3 base	Mortgages/loans	0.7190	0.6977	0.7082	1,826
Qwen3 base	Debt collection	0.8115	0.3262	0.4654	2,244

Table S14 continued

System	Class	Precision	Recall	F1	Support
Qwen3 QLoRA	Retail banking	0.8491	0.8894	0.8686	1,293
Qwen3 QLoRA	Credit reporting	0.9054	0.9294	0.9173	8,571
Qwen3 QLoRA	Credit card	0.7961	0.7550	0.7748	1,494
Qwen3 QLoRA	Mortgages/loans	0.8728	0.8282	0.8497	1,826
Qwen3 QLoRA	Debt collection	0.7769	0.7347	0.7549	2,244
Llama 3.1 base	Retail banking	0.4667	0.7471	0.5745	1,293
Llama 3.1 base	Credit reporting	0.8225	0.8698	0.8455	8,571
Llama 3.1 base	Credit card	0.5323	0.7068	0.6072	1,494
Llama 3.1 base	Mortgages/loans	0.7326	0.6347	0.6802	1,826
Llama 3.1 base	Debt collection	0.8501	0.2754	0.4160	2,244
Llama 3.1 QLoRA	Retail banking	0.8747	0.8806	0.8776	1,293
Llama 3.1 QLoRA	Credit reporting	0.9090	0.9313	0.9200	8,571
Llama 3.1 QLoRA	Credit card	0.7974	0.7784	0.7876	1,494
Llama 3.1 QLoRA	Mortgages/loans	0.8791	0.8227	0.8499	1,826
Llama 3.1 QLoRA	Debt collection	0.7781	0.7540	0.7656	2,244
BERT	Retail banking	0.8463	0.8538	0.8500	1,293
BERT	Credit reporting	0.8985	0.9242	0.9111	8,571
BERT	Credit card	0.7630	0.7517	0.7573	1,494
BERT	Mortgages/loans	0.8336	0.8010	0.8169	1,826
BERT	Debt collection	0.7528	0.6977	0.7242	2,244
DistilBERT	Retail banking	0.8451	0.8438	0.8444	1,293
DistilBERT	Credit reporting	0.8967	0.9216	0.9089	8,571
DistilBERT	Credit card	0.7693	0.7481	0.7583	1,494
DistilBERT	Mortgages/loans	0.8380	0.7950	0.8159	1,826
DistilBERT	Debt collection	0.7420	0.7080	0.7246	2,244
RoBERTa	Retail banking	0.8533	0.8358	0.8444	1,293
RoBERTa	Credit reporting	0.9035	0.9168	0.9101	8,571
RoBERTa	Credit card	0.7549	0.7459	0.7501	1,494
RoBERTa	Mortgages/loans	0.8236	0.8131	0.8182	1,826
RoBERTa	Debt collection	0.7400	0.7198	0.7296	2,244
TF-IDF/LR 10k	Retail banking	0.8421	0.8500	0.8460	1,293
TF-IDF/LR 10k	Credit reporting	0.8822	0.9285	0.9047	8,571
TF-IDF/LR 10k	Credit card	0.7747	0.7135	0.7429	1,494
TF-IDF/LR 10k	Mortgages/loans	0.8263	0.8023	0.8141	1,826
TF-IDF/LR 10k	Debt collection	0.7803	0.6791	0.7262	2,244
TF-IDF/LR full	Retail banking	0.8604	0.8863	0.8731	1,293
TF-IDF/LR full	Credit reporting	0.9192	0.9316	0.9254	8,571
TF-IDF/LR full	Credit card	0.8089	0.8019	0.8054	1,494
TF-IDF/LR full	Mortgages/loans	0.8506	0.8543	0.8525	1,826
TF-IDF/LR full	Debt collection	0.8095	0.7553	0.7815	2,244

Table S15: Observed off-label outputs among 15,428 internal predictions. Compliance is not constrained decoding.

System	Variant	Off-label count
Qwen3 base	base	1
Qwen3 QLoRA	42	0
Qwen3 QLoRA	43	0
Qwen3 QLoRA	44	0
Llama 3.1 base	base	1
Llama 3.1 QLoRA	42	1
Llama 3.1 QLoRA	43	1
Llama 3.1 QLoRA	44	1
BERT	42	0
BERT	43	0
BERT	44	0
DistilBERT	42	0
DistilBERT	43	0
DistilBERT	44	0
RoBERTa	42	0
RoBERTa	43	0
RoBERTa	44	0
TF-IDF/LR 10k	42	0
TF-IDF/LR full	42	0

S4 Benchmark overlaps and per-view descriptions

The twelve files contain 12,000 appearances but only 5,909 unique (label, normalised visible narrative) units. Full-text/prefix linkage forms 3,919 conservative dependency groups. A group draw retains all linked short/long views, with total weight one for repeated identical units. This estimand differs from an unweighted mean of twelve file scores. The files are dependent, shared-source evaluation views. Statistical inference resamples their linked groups rather than treating file scores as independent observations.

Table S16: Benchmark file key. ID order matches both overlap matrices. Imbalanced counts are credit reporting/debt/loans/card/banking; sampling code is unavailable.

ID	Seed ID	Sampling	Chars	Class counts
1	1	balanced	250	200/class
2	1	balanced	500	200/class
3	1	imbalanced	250	511/182/107/100/100
4	1	imbalanced	500	511/182/107/100/100
5	2	balanced	250	200/class
6	2	balanced	500	200/class
7	2	imbalanced	250	511/182/107/100/100
8	2	imbalanced	500	511/182/107/100/100
9	3	balanced	250	200/class
10	3	balanced	500	200/class
11	3	imbalanced	250	511/182/107/100/100
12	3	imbalanced	500	511/182/107/100/100

Table S17: Normalised narrative overlap. Counts are shared unique inputs; diagonals count within-file unique inputs. Row fractions and Jaccard matrices are in the archive.

ID	1	2	3	4	5	6	7	8	9	10	11	12
1	999	490	688	348	1	0	315	142	1	0	3	0
2	490	1000	348	689	0	0	142	311	0	0	0	0
3	688	348	997	521	200	113	5	0	111	60	3	0
4	348	689	521	1000	113	200	0	0	60	111	0	0
5	1	0	200	113	1000	465	180	87	0	0	344	143
6	0	0	113	200	465	1000	87	178	0	0	143	343
7	315	142	5	0	180	87	999	502	90	48	2	0
8	142	311	0	0	87	178	502	1000	48	89	0	0
9	1	0	111	60	0	0	90	48	1000	478	146	79
10	0	0	60	111	0	0	48	89	478	1000	79	146
11	3	0	3	0	344	143	2	0	146	79	998	507
12	0	0	0	0	143	343	0	0	79	146	507	1000

Table S18: Normalised label–narrative-unit overlap. Counts are shared unique inputs; diagonals count within-file unique inputs. Row fractions and Jaccard matrices are in the archive.

ID	1	2	3	4	5	6	7	8	9	10	11	12
1	999	490	688	348	1	0	315	142	1	0	3	0
2	490	1000	348	689	0	0	142	311	0	0	0	0
3	688	348	997	521	200	113	5	0	111	60	3	0
4	348	689	521	1000	113	200	0	0	60	111	0	0
5	1	0	200	113	1000	465	180	87	0	0	344	143
6	0	0	113	200	465	1000	87	178	0	0	143	343
7	315	142	5	0	180	87	999	502	90	48	2	0
8	142	311	0	0	87	178	502	1000	48	89	0	0
9	1	0	111	60	0	0	90	48	1000	478	146	79
10	0	0	60	111	0	0	48	89	478	1000	79	146
11	3	0	3	0	344	143	2	0	146	79	998	507
12	0	0	0	0	143	343	0	0	79	146	507	1000

Table S19: Per-file macro-F1: llm. Multi-seed columns are means. Descriptive scores are not independent replicates or the unique-unit aggregate.

ID	Qwen3 base	Qwen3 QLoRA	Llama 3.1 base	Llama 3.1 QLoRA
1	0.5964	0.7908	0.5839	0.7995
2	0.5923	0.7937	0.5802	0.8029
3	0.6106	0.7891	0.5954	0.8013
4	0.6051	0.7950	0.5898	0.8068
5	0.6222	0.8003	0.6147	0.8037
6	0.6191	0.8041	0.6160	0.8100
7	0.6134	0.8007	0.5683	0.8041
8	0.6210	0.8078	0.5741	0.8123
9	0.6171	0.8016	0.6042	0.8050
10	0.6172	0.8007	0.6058	0.8121
11	0.6360	0.8181	0.5859	0.8155
12	0.6287	0.8234	0.5974	0.8282

Table S20: Per-file macro-F1: baseline. Multi-seed columns are means. Descriptive scores are not independent replicates or the unique-unit aggregate.

ID	BERT	DistilBERT	RoBERTa	TF-IDF/LR 10k	TF-IDF/LR full
1	0.7652	0.7593	0.7603	0.7709	0.8161
2	0.7729	0.7632	0.7640	0.7770	0.8246
3	0.7532	0.7615	0.7542	0.7818	0.8303
4	0.7682	0.7661	0.7575	0.7799	0.8291
5	0.7647	0.7599	0.7701	0.7474	0.8088
6	0.7695	0.7662	0.7677	0.7490	0.8153
7	0.7809	0.7622	0.7628	0.7593	0.7988
8	0.7918	0.7766	0.7762	0.7805	0.8225
9	0.7660	0.7743	0.7814	0.7533	0.8075
10	0.7746	0.7756	0.7829	0.7566	0.8074
11	0.7850	0.7765	0.7957	0.7795	0.8214
12	0.7923	0.7861	0.7927	0.7870	0.8289

S5 Input-length conditions, cohorts and sensitivity analyses

Same-checkpoint conditions are 500 characters/512 tokens (A), 500/2,048 (B), and 1,500/2,048 (C). Models are not retrained for C. Every condition uses the same 15,428 internal IDs; the predefined full-narrative > 500 cohort has 5,770 rows in 5,131 groups. A and B token inputs/predictions are identical, with no extra truncation in either. Their metric differences and bootstrap bounds are exactly zero; exact-response reuse is disclosed in Section S2.

The benchmark files supply only their recorded 250/500-character text; they are not used to construct 1,500-character inputs. Short/long maxima before caps are Qwen 207/403 and Llama 224/421 tokens; all fit the stated caps. Every row records requested/retained characters, tokens and hashes. One common paired group draw is used across the length conditions and checkpoint variants. Longer-text effects include negative zero-shot findings, and do not establish robustness outside this task.

Table S21: Every actual length/checkpoint condition on the complete internal test.

Model	Variant	Condition	Accuracy	Macro-F1	Weighted F1
qwen	base	A	0.7352	0.6295	0.7206
qwen	base	B	0.7352	0.6295	0.7206
qwen	base	C	0.7326	0.6224	0.7154
qwen	42	A	0.8717	0.8378	0.8709
qwen	42	B	0.8717	0.8378	0.8709
qwen	42	C	0.8767	0.8440	0.8757
qwen	43	A	0.8696	0.8332	0.8682
qwen	43	B	0.8696	0.8332	0.8682
qwen	43	C	0.8750	0.8407	0.8735
qwen	44	A	0.8654	0.8282	0.8642
qwen	44	B	0.8654	0.8282	0.8642

Table S21 continued

Model	Variant	Condition	Accuracy	Macro-F1	Weighted F1
qwen	44	C	0.8713	0.8360	0.8699
llama	base	A	0.7295	0.6247	0.7177
llama	base	B	0.7295	0.6247	0.7177
llama	base	C	0.7298	0.6227	0.7167
llama	42	A	0.8743	0.8424	0.8738
llama	42	B	0.8743	0.8424	0.8738
llama	42	C	0.8779	0.8466	0.8772
llama	43	A	0.8729	0.8384	0.8721
llama	43	B	0.8729	0.8384	0.8721
llama	43	C	0.8768	0.8438	0.8757
llama	44	A	0.8736	0.8397	0.8728
llama	44	B	0.8736	0.8397	0.8728
llama	44	C	0.8783	0.8464	0.8773

Table S22: Paired C minus B changes for all metrics and both predefined cohorts.

Model	Variant	Cohort	Metric	Difference	95% CI
qwen	base	all	accuracy	-0.0025	[-0.0047, -0.0003]
qwen	base	all	macro_f1	-0.0072	[-0.0107, -0.0036]
qwen	base	all	weighted_f1	-0.0052	[-0.0077, -0.0027]
qwen	FT mean	all	accuracy	0.0054	[0.0037, 0.0073]
qwen	FT mean	all	macro_f1	0.0072	[0.0048, 0.0097]
qwen	FT mean	all	weighted_f1	0.0053	[0.0035, 0.0071]
qwen	base	> 500	accuracy	-0.0068	[-0.0126, -0.0008]
qwen	base	> 500	macro_f1	-0.0181	[-0.0262, -0.0099]
qwen	base	> 500	weighted_f1	-0.0135	[-0.0201, -0.0068]
qwen	FT mean	> 500	accuracy	0.0146	[0.0099, 0.0195]
qwen	FT mean	> 500	macro_f1	0.0156	[0.0105, 0.0209]
qwen	FT mean	> 500	weighted_f1	0.0142	[0.0096, 0.0192]
llama	base	all	accuracy	0.0004	[-0.0022, 0.0030]
llama	base	all	macro_f1	-0.0020	[-0.0058, 0.0018]
llama	base	all	weighted_f1	-0.0010	[-0.0037, 0.0017]
llama	FT mean	all	accuracy	0.0041	[0.0022, 0.0060]
llama	FT mean	all	macro_f1	0.0055	[0.0029, 0.0081]
llama	FT mean	all	weighted_f1	0.0039	[0.0020, 0.0058]
llama	base	> 500	accuracy	0.0010	[-0.0060, 0.0082]
llama	base	> 500	macro_f1	-0.0017	[-0.0102, 0.0068]
llama	base	> 500	weighted_f1	-0.0018	[-0.0091, 0.0055]
llama	FT mean	> 500	accuracy	0.0109	[0.0060, 0.0160]
llama	FT mean	> 500	macro_f1	0.0117	[0.0065, 0.0172]
llama	FT mean	> 500	weighted_f1	0.0106	[0.0057, 0.0157]

S5.1 Auxiliary length-stratum and matched-view analyses

The following analyses use auxiliary models fitted with full-text-grouped allocations and seed-specific fitting selections. These models have model-visible training/evaluation overlap and are excluded from the primary model comparison. Length strata differ in class/content and do not isolate a truncation effect. For 250/500 matching, only one-to-one label/normalised-250-prefix matches are retained before looking at predictions; ambiguous matches are excluded. Repeated short/long pairs share total weight one and linked views are resampled together. This is not verified underlying CFPB record identity. Filtering evaluation groups does not remove exposure incurred during fitting or selection.

Table S23: Auxiliary-model internal macro-F1 by full narrative length. Inputs are capped at 500 characters; model-visible overlap limits interpretation.

Model	Length	Rows	Base	FT mean
Qwen3-8B	251–500	4212	0.6223	0.8417
Qwen3-8B	≤ 250	5446	0.5849	0.7672
Qwen3-8B	> 500	5770	0.6310	0.8506
Llama-3.1-8B	251–500	4212	0.6301	0.8454
Llama-3.1-8B	≤ 250	5446	0.5460	0.7660
Llama-3.1-8B	> 500	5770	0.6317	0.8545

Table S24: Auxiliary-model 250/500-character matched-view results, unique-pair weighted. The source cohort and fitted models differ from the controlled B/C comparison.

Model	Variant	Unique pairs	250	500	Difference [CI]
Qwen3-8B	Base	1982	0.6297	0.6256	-0.0041 [-0.0149, 0.0063]
Qwen3-8B	FT mean	1982	0.8353	0.8438	0.0085 [0.0008, 0.0162]
Llama-3.1-8B	Base	1982	0.6207	0.6307	0.0100 [-0.0024, 0.0225]
Llama-3.1-8B	FT mean	1982	0.8388	0.8462	0.0074 [0.0001, 0.0150]

S6 Controlled latency and measurement limitations

One hundred internal IDs (20 per class) are selected by fixed hashes; warmup uses the first ten fixed validation IDs. Each model/variant/condition has three repetitions of the same 100 IDs in a hash-defined order. Every timed call is a real generate invocation; no memoisation is used. CUDA is synchronised around generation. End-to-end additionally includes tokenisation, transfer, decoding and regex parsing; loading, warmup and disk I/O are excluded. Base and adapters use the same NF4/double/FP16, SDPA and batch 1; adapters are unmerged. Other CUDA compute causes an abort rather than termination of unrelated processes.

Times are conditional on this RTX 5060 Ti and separate variant job blocks. GPU state is retained before/after each job. The jobs are not a randomised crossover and no thermal equivalence is claimed. The balanced sample need not represent the natural workload. Output lengths differ between base/FT, so differences are not isolated adapter overhead. Repetition means are formed within each ID before a 10,000-draw paired group bootstrap (100 distinct groups). p95 below is the descriptive quantile of the 300 timed calls, not a confidence bound. Requests/second is serial batch-one throughput, not concurrent service capacity.

Table S25: All paired FT-three-seed-mean minus base time differences (seconds), using ID means over three repetitions.

Model	Condition	Timing	Difference	95% CI
qwen	A	Generation	0.1224	[0.1166, 0.1284]
qwen	A	E2E	0.1224	[0.1166, 0.1284]
qwen	B	Generation	0.1207	[0.1150, 0.1266]
qwen	B	E2E	0.1207	[0.1150, 0.1267]
qwen	C	Generation	0.1237	[0.1177, 0.1298]
qwen	C	E2E	0.1237	[0.1177, 0.1298]
llama	A	Generation	0.0802	[0.0758, 0.0847]
llama	A	E2E	0.0802	[0.0758, 0.0847]
llama	B	Generation	0.0800	[0.0756, 0.0844]
llama	B	E2E	0.0800	[0.0756, 0.0844]
llama	C	Generation	0.0837	[0.0793, 0.0882]
llama	C	E2E	0.0837	[0.0793, 0.0883]

Table S26: All 24 fresh-call latency conditions (300 timed calls each; total 7,200). Gen. is generation seconds; E2E includes preprocessing/postprocessing.

Model	Variant	Cond.	Gen. mean	E2E mean	E2E median	E2E p95	Input tok.	New tok.	Req./s
qwen	base	A	0.2891	0.2896	0.2785	0.3433	144.15	8.61	3.453
qwen	42	A	0.4119	0.4125	0.4082	0.4805	144.15	9.09	2.424
qwen	43	A	0.4133	0.4138	0.4084	0.4848	144.15	9.05	2.417
qwen	44	A	0.4090	0.4096	0.4058	0.4759	144.15	9.11	2.441
qwen	base	B	0.2888	0.2894	0.2784	0.3435	144.15	8.61	3.456
qwen	42	B	0.4094	0.4100	0.4061	0.4767	144.15	9.09	2.439
qwen	43	B	0.4077	0.4082	0.4027	0.4762	144.15	9.05	2.450
qwen	44	B	0.4114	0.4119	0.4068	0.4800	144.15	9.11	2.428
qwen	base	C	0.2968	0.2974	0.2924	0.3443	172.20	8.58	3.363
qwen	42	C	0.4158	0.4164	0.4058	0.4946	172.20	9.08	2.401
qwen	43	C	0.4244	0.4251	0.4136	0.5072	172.20	9.12	2.353
qwen	44	C	0.4211	0.4217	0.4097	0.4977	172.20	9.12	2.371
llama	base	A	0.2826	0.2831	0.2691	0.3290	162.12	8.76	3.533
llama	42	A	0.3627	0.3632	0.3584	0.4262	162.12	9.03	2.753
llama	43	A	0.3619	0.3624	0.3565	0.4196	162.12	9.02	2.759
llama	44	A	0.3637	0.3642	0.3591	0.4227	162.12	9.07	2.745
llama	base	B	0.2825	0.2830	0.2690	0.3287	162.12	8.76	3.534
llama	42	B	0.3603	0.3608	0.3567	0.4171	162.12	9.03	2.772
llama	43	B	0.3639	0.3644	0.3591	0.4225	162.12	9.02	2.744
llama	44	B	0.3632	0.3637	0.3591	0.4234	162.12	9.07	2.750
llama	base	C	0.2935	0.2940	0.2829	0.3463	190.18	8.79	3.401
llama	42	C	0.3775	0.3780	0.3670	0.4574	190.18	9.07	2.645
llama	43	C	0.3776	0.3781	0.3655	0.4595	190.18	9.06	2.645
llama	44	C	0.3766	0.3771	0.3637	0.4585	190.18	9.08	2.652

S7 Reference [14] encoder-parameter sensitivity

Güllü et al. (ACDSA 2025, DOI 10.1109/ACDSA65407.2025.11166338), Tables III–IV, report BERT LR 2×10^{-5}, three epochs, batch 16, max 128 tokens, and RoBERTa LR 5×10^{-5}, three epochs, batch 16, max 100 tokens. The reference learning-rate/epoch pairs coincide with the winners of the primary validation search. Separate conditions therefore change physical batch from 4/accumulation 4 to 16/accumulation 1 and caps from 512 to 128/100. Their joint effect is measured; it is not attributed to one parameter. The **main RoBERTa condition uses 500 characters/512 tokens**; only this supplementary condition uses 100 tokens.

We retain the fixed 10k/2k membership, 500-character input, same pinned models, PyTorch/FP16 and main weight decay/warmup/scheduler. The source reports TensorFlow for RoBERTa and different split/preprocessing, so this is not an exact replication or a cross-paper score comparison. All six reference-parameter final-epoch refits (seeds 42/43/44) are reported, without choosing a replacement on test scores.

Both reference-parameter tokenizer audits cover 39,428 inputs, with zero tested split collisions and no violations of the frozen dependency groups. The token caps clip 1 BERT and 18 RoBERTa appearances. Tokenisation truncates the narrative inputs before batch collation; padding to a multiple of eight gives a width of at most 104 for the RoBERTa condition while retaining at most 100 real tokens. Training and evaluation use the same truncation and padding procedure. Each of the six reload parity checks exactly matches all 2,000 saved validation predictions.

Table S27: Reference-parameter conditions, macro-F1. Point estimates refer to the same primary-model predictions; intervals belong to the separate supplementary paired analysis.

View	Model	Setting	Macro-F1	95% CI	Seed SD
internal	bert	main	0.8119	[0.8027, 0.8202]	0.0010
internal	bert	ref14	0.8102	[0.8010, 0.8185]	0.0026
internal	roberta	main	0.8105	[0.8015, 0.8189]	0.0039
internal	roberta	ref14	0.8095	[0.8003, 0.8180]	0.0006
benchmark	bert	main	0.7936	[0.7807, 0.8059]	0.0042
benchmark	bert	ref14	0.7943	[0.7813, 0.8063]	0.0070
benchmark	roberta	main	0.7926	[0.7800, 0.8046]	0.0042
benchmark	roberta	ref14	0.7900	[0.7771, 0.8021]	0.0020

Table S28: All supplementary paired reference-setting minus main-setting differences. The seed order is paired and all 12 prediction variants share each supplementary draw.

View	Model	Metric	Difference	95% CI
internal	bert	accuracy	-0.0016	[-0.0037, 0.0004]
internal	bert	macro_f1	-0.0017	[-0.0045, 0.0010]
internal	bert	weighted_f1	-0.0014	[-0.0035, 0.0006]
internal	roberta	accuracy	-0.0007	[-0.0030, 0.0017]
internal	roberta	macro_f1	-0.0010	[-0.0040, 0.0021]
internal	roberta	weighted_f1	-0.0009	[-0.0032, 0.0015]
benchmark	bert	accuracy	0.0011	[-0.0029, 0.0050]
benchmark	bert	macro_f1	0.0007	[-0.0038, 0.0050]
benchmark	bert	weighted_f1	0.0010	[-0.0031, 0.0049]
benchmark	roberta	accuracy	-0.0014	[-0.0058, 0.0032]
benchmark	roberta	macro_f1	-0.0026	[-0.0076, 0.0026]
benchmark	roberta	weighted_f1	-0.0019	[-0.0065, 0.0028]

Table S29: Every supplementary refit, with the same benchmark unique-input weighting.

Model	Seed	View	Accuracy	Macro-F1	Weighted F1
bert	42	internal	0.8531	0.8131	0.8525
bert	42	benchmark	0.8083	0.7943	0.8062
bert	43	internal	0.8529	0.8088	0.8510
bert	43	benchmark	0.8025	0.7873	0.7997
bert	44	internal	0.8514	0.8086	0.8506
bert	44	benchmark	0.8155	0.8013	0.8142
roberta	42	internal	0.8519	0.8097	0.8508
roberta	42	benchmark	0.8042	0.7902	0.8022
roberta	43	internal	0.8516	0.8088	0.8505
roberta	43	benchmark	0.8076	0.7920	0.8054
roberta	44	internal	0.8520	0.8100	0.8519
roberta	44	benchmark	0.8030	0.7879	0.8018

The internal macro-F1 changes are −0.0017 for BERT and −0.0010 for RoBERTa, with intervals spanning zero. Benchmark changes likewise have intervals spanning zero. These findings do not justify test-based replacement of the main baselines. The slightly different displayed main intervals in this separate 12-variant supplementary analysis reflect its resampling run; primary comparisons use the joint 19-variant main analysis.

S8 Reproducibility and auxiliary protocol analyses

S8.1 Artifact map and verification

The reproducibility repository is available at https://github.com/muhammettan28/CFPB_tuning. It contains source/derived datasets, fixed role/selection manifests, analysis code, validation and final prediction files, fitted sparse objects, configurations and per-run code/environment evidence. Recorded SHA-256 values identify the inputs and results. Neural weights, optimizer states and third-party literature are excluded. Exact base-model snapshots must be downloaded separately. Repeating neural fitting requires a separate project checkout with an empty experiments directory and a registry identifying the downloaded snapshots. Execution scripts use project-relative output paths and reject attempts to overwrite completed runs. Saved predictions support result verification without the omitted fitted weights.

Saved-prediction analysis can instead be repeated without neural weights, a GPU or a local model registry. From the repository root, install `scripts/revision/requirements-cpu-revision.txt`, then run:

```
python scripts/revision/replay_reviewer_results.py \
  --output /tmp/cfpb-review-replay
```

The output directory must not exist. This checks frozen data and source hashes, recomputes all main/reference metrics, seed scores, paired intervals, length sensitivity and latency summaries, and compares nine full result tables to reference CSVs at absolute tolerance 10^{-12}. It writes replay outputs to a separate directory. It does not rerun neural inference or recover unavailable provenance. The replay verifies the supplied result inputs and does not require the manuscript preparation files.

Source hashes identify the analysed files; upstream CFPB-to-Kaggle preprocessing and product aggregation are unavailable. Auxiliary fits lack some immutable upstream identities and complete fitted Qwen artifacts, limiting their reconstruction. Primary-comparison identities are listed in Table S4. Execution metadata can contain absolute machine paths.

Table S30: Key paths relative to the repository root. Complete file-name/hash lists are machine-readable.

Evidence	Location
Frozen data protocol	`analysis/revision_v1/protocol.json`
GPU protocol	`configs/gpu_revision_v1.json`
Reference sensitivity	`configs/ref14_encoder_v1.json`

Table S30 continued

Evidence	Location
Actual tokenizer checks	`analysis/gpu_preparation/input_audit_*.json`
Main results and sources	`analysis/gpu_execution/`
Supplementary sensitivity	`analysis/ref14_encoder/`
Descriptive document tables	`analysis/postprocessing/`
CPU sparse searches	`experiments/tfidf_revision_v1/`
Neural predictions/code	`experiments/gpu_revision_v1/`
Reference-condition runs	`experiments/ref14_encoder_v1/`

Table S31: Frozen protocol SHA-256 values.

Artifact	SHA-256
Data protocol	`89fd382208515e946ab2a941d07809cec8b9fc659d554048f008cf2dea225d0f`
Main GPU plan	`1fe25cdd571eee549060a362980a232fa56978021dc015d25823695086d1d0e2`
Reference plan	`2def9919ab37abd189f9035304c967cf8ede8d80338bf7cf3435afbfd256da6d`

S8.2 Auxiliary data-overlap analysis

The auxiliary inference code tokenises an already rendered chat template with default special-token insertion. The primary pipeline uses `add_special_tokens=False`. A CPU reconstruction with the pinned primary tokenizers produces two leading BOS tokens under the default Llama path and one under the explicit path; the two Qwen paths have identical token IDs for the audited prompt. This identifies a serialization difference, rather than an isolated accuracy ablation. Auxiliary upstream model identities are incomplete, and hardware and quantised numerical execution also differ between the implementations.

Auxiliary models fitted under full-normalised-narrative grouping support a diagnostic analysis of dependence under truncated inputs. Their fitting selections vary with seed, so sample SD combines selection and optimisation variation. At 500 characters, 192 shared unique training inputs are associated with 584 internal-test rows; the three Llama fitting selections expose 383/375/390 test rows. These models therefore have model-visible training/evaluation overlap. Their scores are excluded from the primary comparison. Complete fitted Qwen artifacts and immutable upstream identities for these auxiliary models are unavailable. Differences from the primary comparison also involve membership, serialization, model identities and hardware, so they do not isolate the effect of overlap handling.

Table S32: Internal results for auxiliary models with model-visible overlap. FT is mean plus/minus sample SD across fits with varying selections; these scores are excluded from the primary comparison.

Model	Base mF1	FT mF1	Gain	Base wF1	FT wF1
Qwen3-8B	0.6294	0.8350±0.0017	0.2056	0.7202	0.8686±0.0011
Llama-3.1-8B	0.6189	0.8378±0.0026	0.2189	0.7094	0.8705±0.0011

Table S33: Observed points and paired group-bootstrap intervals for the auxiliary models. Conditioning on fitted models does not eliminate training exposure.

Model	Metric	Base	FT mean	Paired gain
Qwen3-8B	accuracy	0.7350 [0.7186, 0.7519]	0.8696 [0.8600, 0.8789]	0.1346 [0.1241, 0.1452]
Qwen3-8B	macro f1	0.6294 [0.6182, 0.6402]	0.8350 [0.8262, 0.8426]	0.2056 [0.1955, 0.2159]
Qwen3-8B	weighted f1	0.7202 [0.7033, 0.7380]	0.8686 [0.8589, 0.8780]	0.1483 [0.1372, 0.1592]
Llama-3.1-8B	accuracy	0.7319 [0.7159, 0.7488]	0.8717 [0.8623, 0.8807]	0.1398 [0.1294, 0.1503]
Llama-3.1-8B	macro f1	0.6189 [0.6079, 0.6292]	0.8378 [0.8289, 0.8455]	0.2189 [0.2096, 0.2288]
Llama-3.1-8B	weighted f1	0.7094 [0.6921, 0.7274]	0.8705 [0.8610, 0.8796]	0.1611 [0.1500, 0.1725]

The auxiliary internal sensitivity excludes whole components touching the fitting or validation records, leaving 14,826 of 15,428 rows. The conservative benchmark sensitivity leaves 11,986 of 12,000 appearances. These

evaluation filters do not remove fitting/selection exposure and do not validate the primary comparison. Within the 14,826-row internal subset, base/adapted macro-F1 is 0.6290/0.8353 for Qwen and 0.6190/0.8381 for Llama. The adapted-model intervals are [0.8265, 0.8430] and [0.8294, 0.8461]; these remain conditional estimates for the auxiliary models. The final full-data sparse macro-F1 of 0.8476 is independently recalculated from the retained predictions; implementation-audit details remain in the machine-readable evidence.

S8.3 Configuration screening for the shared QLoRA settings

Qwen3-8B configuration screening compares six conditions on the full-text-grouped allocation: natural/balanced sampling, two learning rates, ranks 8/16 and one/two epochs. This limited comparison is not an exhaustive hyperparameter search. Natural sampling with two epochs, learning rate 2×10^{-4} and rank 16 has the highest recorded validation macro-F1. The primary model comparison uses these fixed settings for both Qwen and Llama, without separate family-specific optimisation. The sparse and encoder baselines instead receive their stated validation searches on the common fitting allocation. This asymmetry gives those baselines an explicit tuning opportunity; it does not quantify how different search budgets would change the ranking.

The screening fits use the full-text-grouped candidate allocation, whose model-visible overlap is quantified in Section S8.2. Their scores describe configuration screening rather than independent test performance or an optimal configuration. They are excluded from the primary model ranking. Run identities and source hashes identify all six conditions in the search-summary CSVs.

Table S34: Qwen configuration-screening conditions on the full-text-grouped allocation. Validation scores are auxiliary selection evidence, not primary-comparison test estimates.

Condition	Sampling	LR	Rank	Epochs	Macro-F1	Debt F1
Pilot	natural	2e-04	16	1	0.8098	0.6429
Balanced E1	balanced	2e-04	16	1	0.8040	0.6667
Balanced E2	balanced	1e-04	16	1	0.7948	0.6552
Balanced E3	balanced	2e-04	8	1	0.7951	0.6574
Natural 2-epoch	natural	2e-04	16	2	0.8235	0.7033
Balanced 2-epoch	balanced	2e-04	16	2	0.8065	0.6456

The 154,282-row partitioning pool contains 85,537 credit-reporting records (55.44%), 22,503 debt-collection records (14.59%), 18,367 mortgage/loan records (11.90%), 14,943 card records (9.69%) and 12,932 banking records (8.38%). Its median narrative length was 365 characters and its 90th percentile was 1,261; 63.01% fit within 500 characters. These statistics describe the candidate partitioning pool. The 63.98% coverage in the main text instead describes all 162,405 usable source records. Retained and selected class counts are in Table S1; the populations must not be interchanged.

S8.4 Validation scope

Execution verification checks all 88 main-study stages, actual fitting memberships, the three independently rescored encoder searches, 19 main variants in both views, all same-checkpoint length and controlled-latency files, and source-integrity manifests. The reference audit checks all 12 extra stages, six actual refits, six reload parity tests and independent point metrics. Small hardware/API smoke fits (32 train/8 validation; two optimiser steps) are excluded from scientific results. Frozen prediction checks and tests support the reported computations, but do not establish prospective generalisation, semantic independence, an optimal prompt/budget, operational acceptability or regulatory readiness. The reproducibility package provides the source hashes, selections and recorded outputs for the stated verification scope.

S8.5 Computational cost and CPU baseline timings

Table S35 reports implementation-specific training wall times. QLoRA runtimes include in-training validation/checkpointing; encoder Trainer runtimes exclude the separate final validation call. Means use three refits for each neural method. Sparse times are fresh fixed-configuration, seed-42 vectoriser-plus-classifier fits on the CPU; all 15,428 predictions match the retained sparse models. Data loading and validation are

outside the sparse fitting timer. Encoder search totals sum the six recorded training phases and exclude setup, validation and I/O. Sparse search totals are the recorded full search wall times. QLoRA settings are fixed for the primary comparison, so no primary-allocation search time is reported. Device-hours below are elapsed wall hours on the named device, rather than measured GPU busy-hours or energy.

Table S35: Recorded neural training and measured sparse training times. Fit hours are per refit; search hours use the stated method-specific timing boundaries. A dash denotes fixed QLoRA settings, not a measured zero search cost.

System	Fits	Fit device	Mean fit hours	Search hours
Qwen3 QLoRA	3	RTX 5060 Ti	6.9612	—
Llama 3.1 QLoRA	3	RTX 5060 Ti	7.1541	—
BERT	3	RTX 5060 Ti	0.0427	0.2160
DistilBERT	3	RTX 5060 Ti	0.0254	0.1239
RoBERTa	3	RTX 5060 Ti	0.0453	0.2307
TF-IDF/LR 10k	1	CPU	0.0004	0.0218
TF-IDF/LR full	1	CPU	0.0134	0.3217

CPU inference uses Intel(R) Core(TM) i7-14700F, four threads, batch one, ten validation warmups and the same 100 internal IDs and three hash-defined repetition orders as the LLM latency study. Each encoder uses its seed-42 checkpoint in FP32; the sparse classifiers use float64 features and their fixed selected configurations. End-to-end time includes tokenisation/vectorisation, classification and label extraction; loading, warmup and I/O are excluded. The CPU is not reserved exclusively, and measured load averages are retained. The raw 300 calls per system, environment and source hashes are supplied under `experiments/cpu_costs_v1/`. These device/precision-specific timings are descriptive implementation costs; they are not same-device architecture comparisons or service-capacity estimates.

Table S36: Four-thread CPU end-to-end inference on the matched latency cohort (milliseconds). Each row has 300 fresh calls. Median and p95 are empirical call quantiles.

System	Calls	Mean ms	Median ms	p95 ms
BERT	300	39.678	42.432	75.221
DistilBERT	300	19.511	21.117	32.631
RoBERTa	300	35.820	40.846	48.113
TF-IDF/LR 10k	300	0.380	0.344	0.501
TF-IDF/LR full	300	0.293	0.298	0.328